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Interest Rate Swaps

Classical Plain Vanilla Swap

Single-Curve Discounting

Two Parties A, B

Notional Face (never paid)

T - multiple of 6 months

$$T_i = \frac{1}{\alpha}, \quad i=0,1,2, \dots, \alpha T$$

At each T_i ($i \geq 1$) B pays $F \frac{\alpha}{2}$
(fixed interest to A)

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A is called the receiver (of fixed)

A pays B $F \frac{r_{i-1}}{2}$ at each T_i ,
 $i \geq 1$ (Floating Interest)

r_{i-1} Floating rate that prevails
at T_{i-1} for borrowing between
 T_{i-1} and T_i .

g needs to be determined so
that neither pays anything
at time 0

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Typically Collateral must be posted.

Floating Rate Bonds

F, T Get $F \frac{r_{i-1}}{2}$ at T_i for $i=1, 2, \dots, 2T$ plus F at T .

Time-0 Price is F

④ Just prior to each T_i , Price of floater is F_i .

Replicate A's position on Swap

$t=0$ buy Coup F, T, q (Gives fixed pays + F at maturity)

Sell Floater F, T

(Forces me to pay floating and F at T)

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Choose q to make price of
Portfolio = 0 at time 0

$$Fd(\tau) + Fq \sum_{i=1}^{\tau-1} d(\frac{i}{2}) - F = 0$$

$$q = \frac{2[F(1-d(\tau))]}{\sum_{i=1}^{\tau-1} d(\frac{i}{2})}$$

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In practice most swaps in
US pay fixed twice per year
and floating 4 times per
year.

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LIBOR

London Interbank Offered
Rate

Avg. of rates at which banks
will lend money (unsecured) to
other banks.

Many deriv Securities and
variable mortgage rates have
had payments based on LIBOR

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LIBOR was scheduled to
Stop in 2021

1 week USD LIBOR and 2m USD
LIBOR will stop being quoted on
Dec 31, 2021

Others scheduled to stop on
June 30, 2023

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LIBOR was not based on
actual transactions
(LIBOR Fixing Scandal)

One Important Replacement
SOFR (Secured Overnight
Financing Rate)

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LIBOR

Credit Risk
Estimate

Swap Pays
Known full
Period in
advance

SOFR

Secured
Based on Actual
Transaction

SOFR Daily Rate
3-m floating rate
daily rate compounded
for 3 months: Floating
Payment known only
at last minute

LIBOR

5 Currencies

USD, GBP, EUR, JPY, CHF

Most important for US

3m USD LIBOR (.131% today)

SOFR (.05% today)

US 3m T-Bill (.052%)

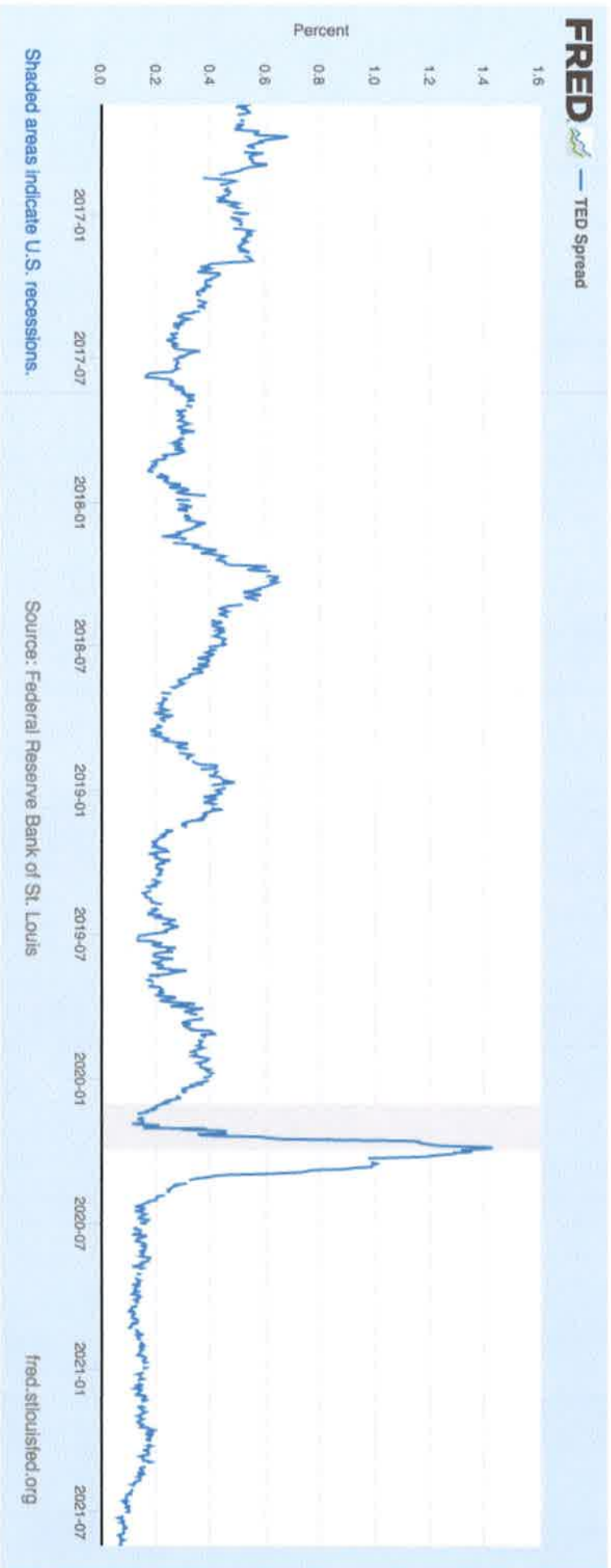
3m LIBOR - 3m T-Bill $\approx$ 8bp Today

TED Spread

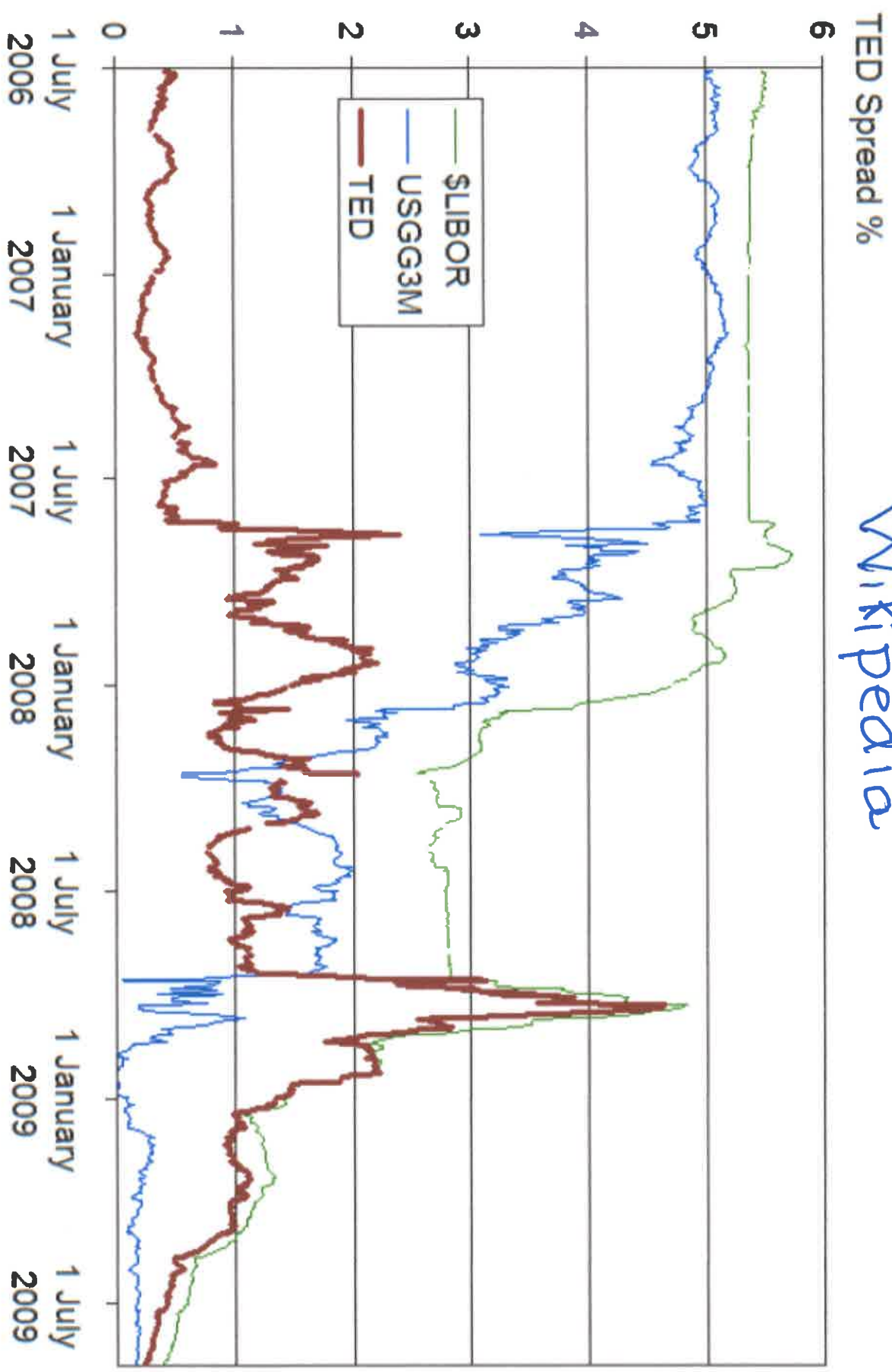
3m LIBOR - 3m TBill

typically between 0,50 bps
(30 bp considered standard)

In 2008, TED exceed 500 bps!



Wikipedia



Eurodollar Futures

Payments Based on

Minus the interest on a
\$1,000,000 3m LIBOR Deposit
(at maturity)

Long Eurodollar Futures
position loses money when
rates rise

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Price Quotation

100(1-L) L annualized Libor

97.6475

Corresponds to

$L = 2.3525\%$

Value of Margin Acct changes
by 2,500 x change in quoted
Price

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DV01 of 1 Eurodollar
Futures Contract is 25

Eurodollar Futures Calendar Spread

Believe Curve will steepen

Buy Short Maturity ED Futures
Sell Long Maturity ED Futures

Ratio (Buy & Sell Same # Contracts)

Principal Components

Use Statistical analysis to identify changes in shape of yield curve that are most significant

3 types of changes typically capture 96-98% variance in yield curve movements

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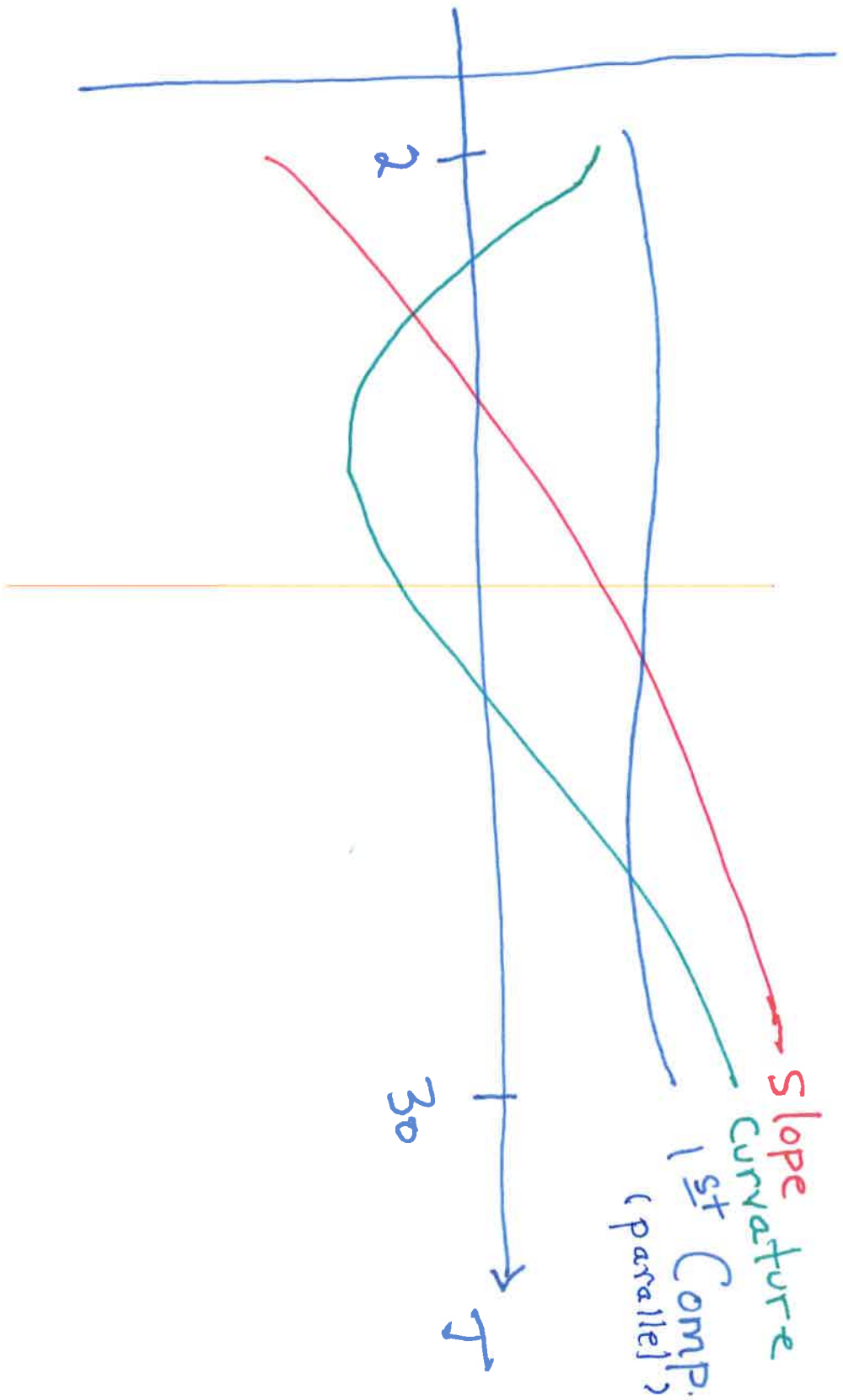
For US

1st Prin. Component (Appx Parallel Shift) Captures $\approx 80\%$ Variance

2nd Prin. Component Slope
1st 2 comps $\approx 90\%$ Variance

3rd Curvature

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Try Yourself

Daily Treasury yields over past
2 to 3 months

$T = 2, 3, 5, 7, 10, 20, 30$

Differences in Daily yields for
each maturity

Covariance Matrix

Eigenvector Prin Comp.

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Largest eval $\rightarrow$ 1st Component
2nd Lgst eval $\rightarrow$ 2nd Component
3rd Lgst eval $\rightarrow$ 3rd Component

1st Comp. Parallel
2nd Comp. Slope
3rd Comp. Curvature

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Option Spread Trades

Sell an Option

Buy a different option

Could pay $y > 0$ to enter

Terminal Capital $> y$ (to profit)

Could Receive $x > 0$ upon entry

Want $x + \text{terminal capital} > 0$

(to profit)

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Vertical Spread

Maturity T Two Strikes $0 < K_1 < K_2$

Buy One Option (Sell another
of Same type [put, call])

Same Maturity Different Strikes

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Bull Spread (Calls)

Buy K_1 -call, Sell K_2 -call

Costs money to enter

Bull Spread (Puts)

Buy K_1 -put Sell K_2 -put

Receive Money to Enter

Bear Call Spread
Bear Put Spread

Exercise

Calendar Spreads

Buy One Option, sell another
of same type, same strike,
but different maturity

$$0 < T_1 < T_2$$

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Bull Calendar Spread (Calls)

Buy T_2 -Call, Sell T_1 -Call

(Usually, K slightly larger
than current stock price)

Costs money to enter

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Is this very risky?

T_2 -call always worth more
than T_1 -call

No Liability after initial
outlay of cash

Trade Could Cost a lot
to enter

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Option Spread Trades

Vertical Spreads

Calendar Spreads

Diagonal Spreads

Require a Directional View

Profit on Volatility

1. Straddle
2. Strangle

Straddle: Buy one call, buy one put, same exp, same strike

Usually $T \approx 1$ month
or less

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Strangle: Buy one call strike K_2

Buy one put strike K_1

$$0 < K_1 < K_2$$

Same T

Costs Less than Straddle

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Q: How do we profit if we believe stock price won't move much?

Suppose we believe that

$K_2 \leq \text{stock price} \leq K_3$
in future

Sell Call K_3 mat T

(won't have to pay if

Stock Price $\leq K_3$)

Sell Put K_2 mat T

(won't have to pay if

Stock Price $\geq K_2$)

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Sold Options Create Large
Potential Liability

Buy call mat T $K_4 > K_3$

Buy put mat T $K_1 < K_2$

Still Receive Money at $t=0$!

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Called Iron Condor

Advice for MSCF

- Balance Between Coursework and Internship Search
(Key 1st Mini Classes helpful for Interviews)

- Know Yourself
- Time Management
- Avoid Time Sinks (Online Distractions, Poker)

- Personal Wellness
- Communication Skills
 - Oral & Written
 - Speak English as much as possible (Use Ed Barr as a Resource)
- Follow Markets
- If you are in trouble, ask for help right away.

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- Be Polite - Don't Burn Bridges
- Network (but don't become annoying)
- Listen to Career Services
Vaishali, Ashley, Queeta, Daniel
- Listen to Academic Advisors
(Diffy & Jess)
- Cooperate with one another
- Include BSCF students in
Networking